Property (OB) Need Not Pass to Open Subgroups

Candidate counterexample packet for arXiv:1203.6047
Model: GPT5.6

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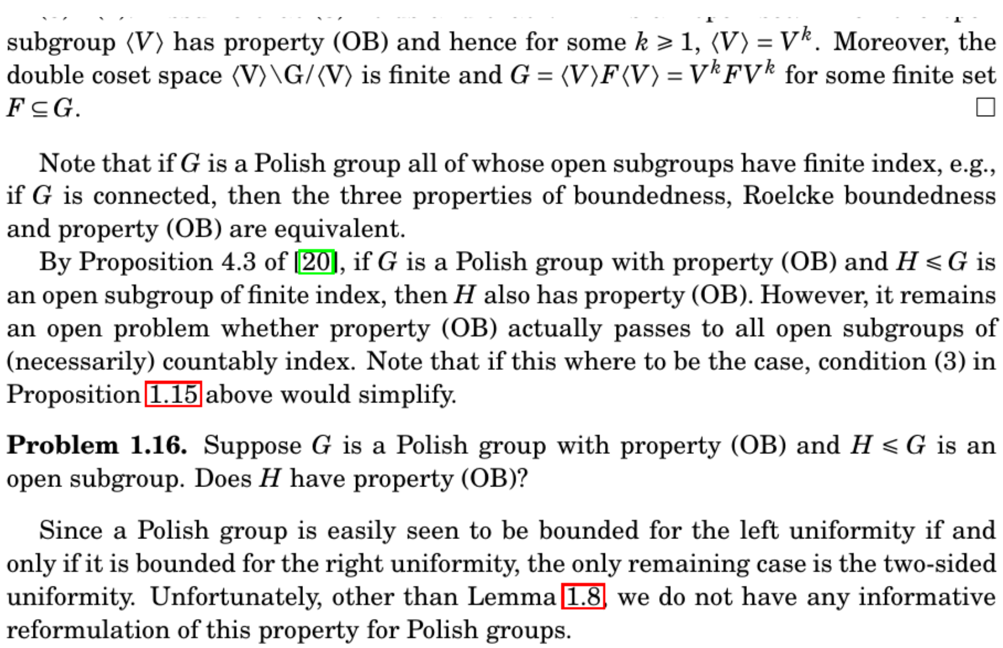
Status. `candidate_counterexample_likely_valid_novelty_unconfirmed`.

1 Source problem

Christian Rosendal asks the following in Problem 1.16 of *Global and local boundedness of Polish groups* (PDF page 14 of the arXiv version) [1]:

> Suppose G is a Polish group with property (OB) and $H \leq G$ is an open subgroup. Does H have property (OB)?



subgroup $\langle V\rangle$ has property (OB) and hence for some $k \geqslant 1$, $\langle V\rangle = V^k$. Moreover, the double coset space $\langle V\rangle\backslash G/\langle V\rangle$ is finite and $G = \langle V\rangle F\langle V\rangle = V^k F V^k$ for some finite set $F \subseteq G$. □

Note that if G is a Polish group all of whose open subgroups have finite index, e.g., if G is connected, then the three properties of boundedness, Roelcke boundedness and property (OB) are equivalent.

By Proposition 4.3 of [20], if G is a Polish group with property (OB) and $H \leqslant G$ is an open subgroup of finite index, then H also has property (OB). However, it remains an open problem whether property (OB) actually passes to all open subgroups of (necessarily) countably index. Note that if this where to be the case, condition (3) in Proposition 1.15 above would simplify.

Problem 1.16. Suppose G is a Polish group with property (OB) and $H \leqslant G$ is an open subgroup. Does H have property (OB)?

Since a Polish group is easily seen to be bounded for the left uniformity if and only if it is bounded for the right uniformity, the only remaining case is the two-sided uniformity. Unfortunately, other than Lemma 1.8, we do not have any informative reformulation of this property for Polish groups.

Figure 1: The source question and its immediate context, arXiv:1203.6047, PDF page 14.

For a Polish group K, we use Rosendal's equivalent bounded-width characterization [1, Theorem 1.11]:

$$K \text{ has property (OB)} \iff \begin{array}{l}\text{for every nonempty symmetric open } V \subseteq K \text{, there are a finite}\\ F \subseteq K \text{ and } k \geq 1 \text{ such that } K = (FV)^k.\end{array} \tag{1}$$

Equivalently, property (OB) fails if the group admits an unbounded continuous length function [1, Theorem 1.11].

2 Construction and proof intuition

Let $P = S_\infty$ be the full symmetric group of $\mathbb{N}$, with the topology of pointwise convergence, and let $N = \mathbb{Z}^\mathbb{N}$ have the product topology. The group P acts continuously on N by permuting coordinates,

$$(p \cdot a)_i = a_{p^{-1}(i)}.$$

Set $G = N \rtimes P$.

The key idea is that a neighbourhood of the identity only constrains finitely many coordinates. A single finite permutation moves those constrained coordinates to a disjoint block, where arbitrary values are allowed. This gives bounded width for the whole base group N. The permutation part also has bounded width because the pointwise stabilizer of a finite set has only finitely many double cosets in P. Together these observations verify (1) for G.

On the other hand, once the permutation part is required to fix coordinate 0, projection onto that coordinate becomes a homomorphism onto $\mathbb{Z}$. The absolute value of this homomorphism is an unbounded continuous length function. Thus the open coordinate stabilizer fails property (OB).

3 Formal counterexample

We use additive notation on N. The multiplication in G is

$$(a, p)(b, q) = (a + p \cdot b, pq).$$

For a finite set $A \subseteq \mathbb{N}$, define

$$N_A = \{a \in N : a_i = 0 \text{ for all } i \in A\}, \qquad P_A = \{p \in P : p(i) = i \text{ for all } i \in A\},$$

and $U_A = N_A \rtimes P_A$.

Lemma 1. *The subgroups U_A, with $A \subseteq \mathbb{N}$ finite, form a neighbourhood basis at the identity of G.*

Proof. Since P_A fixes every coordinate in A, it preserves N_A, so U_A is a subgroup. A basic identity neighbourhood in the product topology constrains only finitely many coordinates of N and requires the permutation to fix only finitely many indices. Enlarging the union of these two finite sets to A gives U_A contained in that neighbourhood. □

Lemma 2. *For finite $A \subseteq \mathbb{N}$, the double-coset space $P_A \backslash P / P_A$ is finite.*

Proof. To $p \in P$, associate the partial injection

$$\theta_p : \{i \in A : p(i) \in A\} \longrightarrow A, \qquad \theta_p(i) = p(i).$$

Left or right multiplication by an element of P_A does not change θ_p.

Conversely, suppose $p, q \in P$ induce the same partial injection. On the indices $i \in A$ sent outside A, the points $p(i)$ and $q(i)$ form two equally sized tuples of distinct points in $\mathbb{N} \setminus A$. Choose $u \in P_A$ with $u(p(i)) = q(i)$ for all such i. On the remaining indices of A, equality of the partial injections and the fact that u fixes A give $up(i) = q(i)$. Hence $(up)^{-1}q$ fixes A pointwise; writing this permutation as $v \in P_A$ gives $q = upv$. Thus the double cosets are classified by partial injections of the finite set A into itself, of which there are only finitely many. □

Theorem 3. *There is a Polish group with property* (OB) *having an open subgroup without property* (OB)*. In particular, the answer to Rosendal's Problem 1.16 is negative.*

Proof. Both $N = \mathbb{Z}^{\mathbb{N}}$ and $P = S_\infty$ are Polish, and the coordinate-permutation action is continuous. Hence $G = N \rtimes P$ is Polish.

We first prove that G has property (OB). Fix an identity neighbourhood $W \subseteq G$. By Lemma 1, choose finite A with $U_A \subseteq W$. Choose a finite set $B \subseteq \mathbb{N} \setminus A$ with $|B| = |A|$, and let $\tau \in P$ be an involution exchanging A and B and fixing the remaining coordinates.

For $a \in N$, let $x \in N_A$ agree with a on $\mathbb{N} \setminus A$ and vanish on A. Then $a - x$ is supported on A. Put $y = \tau^{-1} \cdot (a - x) = \tau \cdot (a - x)$; this is supported on B and hence belongs to N_A. Directly from the multiplication law,

$$(a, 1) = (x, 1)(0, \tau)(y, 1)(0, \tau). \tag{2}$$

By Lemma 2, choose a finite set $E \subseteq P$ of representatives for $P_A \backslash P / P_A$. For every $p \in P$, write $p = uev$ with $u, v \in P_A$ and $e \in E$. Combining this with (2) shows

$$(a, p) \in U_A(0, \tau)U_A(0, \tau)U_A(\{0\} \times E)U_A.$$

If

$$F = \{(0, 1), (0, \tau)\} \cup (\{0\} \times E),$$

then

$$G = (FU_A)^4 \subseteq (FW)^4. \tag{3}$$

To match the exact form of (1), let $V \subseteq G$ be any nonempty symmetric open set. Then V^2 is an identity neighbourhood. Apply (3) with $W = V^2$ to get $G = (FV^2)^4$. After adjoining the identity to F, each factor fv_1v_2 splits as $(fv_1)(1v_2)$, and therefore $G = (F'V)^8$ for a finite F'. Criterion (1) proves that G has property (OB).

Now let $P_0 = P_{\{0\}}$ and

$$H = N \rtimes P_0.$$

This is an open subgroup of G. Define

$$\varphi : H \longrightarrow \mathbb{Z}, \qquad \varphi(a, p) = a_0.$$

If $p \in P_0$, then $(p \cdot b)_0 = b_{p^{-1}(0)} = b_0$, so the multiplication law gives

$$\varphi((a, p)(b, q)) = a_0 + b_0 = \varphi(a, p) + \varphi(b, q).$$

Thus φ is a continuous surjective homomorphism. Consequently $\ell(a, p) = |a_0|$ is an unbounded continuous length function on H. Rosendal's length-function characterization implies that H does not have property (OB). $\square$

4 Verification notes

The proof is non-computational. The following points were checked directly.

- The action $P \curvearrowright N$ is continuous in the stated Polish topologies.
- U_A is a subgroup because P_A preserves the condition of vanishing on A.
- Lemma 2 proves, rather than assumes, the finite double-coset assertion.
- Expanding the semidirect-product law in (2) yields $x + \tau \cdot y = x + (a - x) = a$ and permutation part $\tau^2 = 1$.
- The four blocks in (3) are $(1U_A)((0, \tau)U_A)((0, \tau)U_A)((0, e)U_A)$.
- The map φ is a homomorphism exactly because all permutation parts in H fix 0.

5 Scope and novelty status

This construction completely answers the yes/no source problem if the proof is accepted. It does not claim a classification of those property-(OB) Polish groups whose open subgroups retain property (OB).

A bounded novelty search on 11 August 2026 checked the run's registry, solution, attempt, and proof-gap indexes and searched exact source wording, both problem labels, and combinations of "property (OB)", "open subgroup", "counterexample", "wreath product", and the displayed semidirect product. No later paper claiming an answer was found. This was not an exhaustive citation search, so novelty remains unconfirmed and should be checked by a specialist before any public originality claim.

6 Human-review recommendation

High priority. The central audit is the bounded-width factorization proving property (OB) for $\mathbb{Z}^{\mathbb{N}} \rtimes S_\infty$. The rest of the obstruction is the explicit quotient map onto $\mathbb{Z}$.

References

[1] Christian Rosendal, *Global and local boundedness of Polish groups*, Indiana Univ. Math. J. **62** (2013), no. 5, arXiv:1203.6047, doi:10.1512/iumj.2013.62.5133.